

DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (AI) IN MASTER FINAL PROJECT (TFM)

I have used Generative AI in my TFM

Check all that apply:

YES ☐ **NO** ☐

If you have ticked YES, please complete the following 3 parts of this document:

Part 1: Reflection on legal, ethical and responsible behavior

Please be aware that the use of AI carries some risks and may lead to a series of serious academic consequences: the TFM will not be evaluated by the University if the use of AI involves the use of confidential data, copyrighted materials, or personal data, used without complying with the conditions required in each case (authorization of the interested parties, authorization of the owners, following the instructions of the University).

Question	
1. In my interactions with AI tools, I have provided confidential data, always with the appropriate authorization of the data subjects. Confidentiality encompasses any information that a person or organization wishes to protect for legal, commercial, privacy, or strategic reasons (such as patents or trade secrets).	
YES, I have used this data with permission	NO, I have not used confidential data
2. In my interaction with Generative AI tools, I have submitted copyrighted materials with the permission of those concerned.	
YES, I have used these materials with permission from the copyright holders, or without permission because they fall within one of the exceptions or limitations permitted by law: • Public domain work • Licensed work (Creative Commons licenses) • Use of excerpts for research purposes (fair use)	NO, I have not used copyrighted materials

3. In my interaction with Generative AI tools, I have submitted personal data with the consent of the data subjects.	
YES, I have used this data with the authorization of the interested parties and in accordance with the instructions contained in the guide approved by the University .	NO, I have not used personal data
4. My use of the Generative AI tool has respected its terms of use , as well as the essential ethical principles, not being maliciously oriented to obtain an inappropriate result for the work presented, that is to say, one that produces an impression or knowledge contrary to the reality of the results obtained, that supplants my own work or that could harm people.	
YES	NO

Part 2: Declaration of technical use

Documentation and drafting:

- *Supporting reflection in relation to the development of the work: iterative process of analysing alternatives and approaches using AI*

I declare that I have made use of the Generative AI system **Gemini Flash 3.6** for **documentation and drafting**, specifically to help me understand, on several occasions, the content of the official technical sources underpinning this work, where that content was ambiguous or difficult to process, so that I could address those ambiguities appropriately in my own writing, using the prompt: *"I am reading the official Tor specification on the shared random value and on onion-service descriptor rotation. It says the shared random value is produced once per 1440-minute period and that descriptors are re-uploaded at time-period boundaries. Can you explain, in plain terms, how the shared random value's daily refresh and the descriptor's time-period rotation relate to each other, and whether they happen on the same schedule?"*, having as interaction the clarification of the relationship between the 24-hour shared-random-value refresh and the descriptor time-period rotation.

- *Revision or rewriting of previously drafted paragraphs*

I declare that I have made use of the Generative AI system **Duck AI, with model GPT-5.6 Luna**, for the **revision or rewriting of previously drafted paragraphs**, to help adapt paragraphs I had already written into the formal register expected of an academic research paper. I requested complete rewrites of passages whose information was particularly hard to adapt to that register, and I also used it to obtain a suitable synonym for a given word by supplying a complete sentence as context and asking for a synonym that fit that context, using the prompt: *"Rewrite the following paragraph so that it reads in the formal, precise style of a security research paper, keeping all technical facts unchanged: [paragraph I had already drafted]"*, having as interaction the adoption of a revised wording of that paragraph, which I reviewed and edited before incorporating it and, separately, using the prompt: *"In the sentence '[complete sentence]', suggest a more precise synonym for the word '[word]' that fits this context"*, having as interaction the selection of the proposed synonym where it genuinely fit the sentence.

- *Search for information or answers to specific questions*

I declare that I have made use of the Generative AI system **Gemini Flash 3.6** for the **search for information or answers to specific questions**, to answer specific questions I had about the official information on Tor when that information was ambiguous or difficult to comprehend, using the prompt: *"According to Tor's official specifications, at what UTC time is the new shared random value published each day, and how long before that boundary should a measurement tool stop trusting the current period's descriptor positions to avoid mixing two time periods?"*, having as interaction the answer regarding the 00:00 UTC period boundary and the practical implication of an exclusion window around it.

- *Bibliography search*

I declare that I have made use of the Generative AI system **Claude Opus 4.8 (High effort)** for **bibliography search**, to locate specific, verifiable, peer-reviewed studies dealing precisely with Hidden Service Directories (HSDirs). This was necessary because research addressing exactly what this work measures is scarce: most available literature concerns onion services in general rather than the directory relays themselves, which made finding sufficiently specific related work difficult, using the prompt: *"Find related-work studies on Tor V3 HSDir relay nodes for a thesis bibliography. This is a measurement study of the HSDir relays themselves, not of the onion services (the onion services are only the probe used to discover which HSDirs are active and where they sit in the hash ring). Find 3–4 real, verifiable, peer-reviewed studies for each of these objectives: uptime and stability of v3 HSDir nodes, churn and sybil patterns of v3 HSDirs, anomalous HSDir behaviour (nodes appearing, disappearing, or changing, and whether a failure is node-level or operator-level), related areas such as hash-ring position observation, sybil detection via IP ranges/family, and operator fingerprinting. Only reputable venues or the Tor Project's own research. Every entry needs a working link or DOI. Never invent a reference"*, having as interaction the identification of candidate studies, each with a verifiable DOI or link and a justification of its relevance, which I then reviewed, verified independently.

- *Summary of bibliography consulted - N/A*
- *Translation of texts consulted - N/A*

Develop specific content

Generative AI has been used as a support tool for the development of the specific content of the TFM, including:

- *Assistance in the development of lines of code (programming)*

I declare that I have made use of the Generative AI systems **Gemini Flash 3.6**, **Claude Sonnet 5 (Medium effort)**, and, on occasions where the logic of the analyses was of particularly high complexity, **Claude Opus 4.8 (High effort)**, for **assistance in the development of lines of code (programming)**. The programming logic and design were my own, the AI tools implemented that logic under my direction and supervision. The data-collection tool which interacts with Tor's ControlPort over a concurrent, five-thread architecture (a network-wide consensus auditor, a tracked-HSDir consensus monitor, an active descriptor prober issuing lookups, an asynchronous event listener, and a scaled ephemeral-onion manager), with locks to prevent race conditions during continuous logging was developed with the assistance of **Gemini Flash 3.6**. The analysis program, which comprises a large set of analysis functions (uptime and stability, churn and flag dynamics, sybil concentration via family-identifier and IP/subnet clustering, hash-ring position and coverage, and descriptor upload/fetch hosting), was developed with the assistance of **Claude Sonnet 5 (Medium effort)**, escalating to **Claude Opus 4.8 (High effort)** for the higher-complexity analysis logic.

- *Generation of diagrams, images, audios or videos - N/A*
- *Optimisation processes*

I declare that I have made use of the Generative AI systems **Gemini Flash 3.6** and **Claude Sonnet 5 (Medium effort)** for **optimisation processes**. I used **Gemini Flash 3.6** to optimise the data-collection processes of the fetcher, removing inconsistencies and unnecessary steps from the collection pipeline. I used **Claude Sonnet 5 (Medium effort)** to optimise several of the analysis processes, given the considerable computational effort required to produce the results presented in this work for example, replacing slower row-by-row processing with vectorised operations so that a month of hourly data could be analysed efficiently.

- *Data processing: collection, analysis, cross-checking of data...*

I declare that I have made use of the Generative AI systems **Gemini Flash 3.6**, **Claude Sonnet 5 (Medium effort)**, and, where the logic of the analyses was of particularly high complexity, **Claude Opus 4.8 (High effort)**, for **data processing (collection, analysis, and cross-checking of data)**. I used **Gemini Flash 3.6** to help build the information fetcher, and **Claude Sonnet 5 (Medium effort)** and **Claude Opus 4.8 (High effort)** to help build the analysis program. I also used these tools to cross-check the processing against a small training/example dataset with known characteristics, to confirm that the computed results and the generated visualisations were correct and internally consistent, and to help resolve any inconsistencies found in that check, using the prompt: "Add a complete Python function to `hmdir-analysis.py` that measures sybil behaviour among HSDir relays. Given a dataframe and a context label, for each hourly snapshot group the relays by their family identifier (`declared_family_id`), and for each family count active relays (status `ACTIVE_IN_RING`) versus inactive relays. Classify the operator as a partial drop if some of its relays are inactive while others remain active, and a total drop if all of its relays are inactive; also compute the network-share of relays dropped per snapshot. Plot per-day absolute partial/total drop counts and their network-shares, save the figure under `analysis-results/sybil/family-id/`, and print the top-5 families by each drop type. If the context label is 'tracked', additionally repeat the whole analysis per service type. Exclude the 'None' (no-family) bucket by default. Use only non-deprecated pandas/matplotlib APIs", having as interaction the incorporation of the resulting function into the analysis program, which I then reviewed, ran against the sample data, and verified before use.

- *Inspiration of ideas in the creative process - N/A*
- *Other uses linked to the generation of specific points of the specific development of the paper*

I declare that I have made use of the Generative AI systems **Claude Sonnet 5 (Medium effort)** and **Claude Opus 4.8 (High effort)** for **other uses linked to the generation of specific points of the specific development of the paper**, as a support to the analysis of the results rather than as a substitute for it. After producing the results and their visualisations, I directed a structured re-examination of them. I posed specific questions and specified particular relationships to verify across the different analyses, and the tools carried out those checks and reported back what they found. This directed cross-checking helped in two ways. First, it surfaced connections between results across different analyses for instance, that the third-party-probe population behaved distinctly on several separate metrics for one shared underlying reason, which were consistent in the data but which I had not immediately identified on a first pass. Second, it helped confirm results that were laborious to derive by hand and to flag apparent inconsistencies between an analysis and its visualisation for me to investigate and resolve. In every case the interpretation, the decision on what to verify, and the final conclusions were my own. The tools executed the rechecks I specified and returned their findings for my assessment.

Part 3: Reflection on utility

Please provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of Generative AI tools in the development of your work. Mention if it has helped you in the learning process, or in the development or drawing conclusions from your work.

The use of Generative AI tools was, on balance, a clear net benefit to this work, primarily as a comprehension aid and as an implementation accelerator under my direction, rather than as a replacement for my own work.

The most valuable contribution was to my own understanding. Tor's V3 onion-service protocol is documented in dense, highly technical official specifications, and being able to pose targeted questions about ambiguous or hard-to-process passages, which let me arrive at a precise and unambiguous understanding of the material, of the scope of the research, and of what was actually feasible given that material, much faster than working from the raw specifications alone. This clearer comprehension led directly into a better-scoped project and more accurate writing.

The tools were also a substantial help in building the project's technical infrastructure. The research required a considerable amount of engineering before any analysis could begin, a custom, continuously running data collector built on a five-thread concurrent architecture over Tor's ControlPort, a full month of uninterrupted hourly data collection, and then a broad analysis program spanning uptime and stability, churn and flag dynamics, sybil concentration via family and subnet clustering, hash-ring position and coverage, and descriptor upload/fetch hosting. The logic and design of these tools were my own contribution, the AI implemented that logic under my supervision, which let me move faster through a large implementation surface, obtain useful feedback, and resolve issues and inconsistencies more quickly than I could have alone. Given the fixed one-month collection window, time saved on implementation was time that could go toward analysing, correlating, and interpreting the results. Finally, using AI to cross-check results and visualisations against a known example, and to help discover connections across analyses, improved my confidence in the correctness and consistency of the findings.

Nonetheless the tools were only as useful as the direction I gave them. They required precise, well-specified instructions, and their output always needed my review, verification, and correction before it could be trusted or used. Outputs were not reliable enough to accept unchecked, particularly for the higher-complexity analysis logic and for any factual claim about the protocol, which I verified against the official sources. Model and version behaviour also varied, and more capable models were needed for the harder logic. Crucially, the tools did not do the work for me, they did not choose the research question, design the measurement methodology, decide what to measure or why, or draw the conclusions. Those remained my own responsibility. Used with that understanding, Generative AI allowed me to deliver the full scope and infrastructure of this research to a higher standard and in less time, and it genuinely supported my learning throughout.